

ADHEESH TRIVEDI

✉ adheeshtrivedi@gmail.com | 💻 adhxytry | 🔗 AdhTri001 | 🌐 adhxytry.github.io

4TH YEAR, BS COMPUTER SCIENCE & ENGINEERING, IISER BHOPAL, INDIA

EDUCATION

IISER Bhopal | GPA: 9.08^{*}/10 (3.63/4)

2022 – 2026

B.S. | Computer Science & Engineering

Bhopal, India

O (Outstanding): 10/10 • A: 10/10 • B+: 9/10 • B: 8/10

Multivariable Calculus (B+)
Discrete Mathematics (A)
Probability & Statistics (A)
Complex Variables (A)
Real Analysis (B+)
Applied Optimization (B+)
Signals & Systems (A)

Theory of Computation (A)
Principles of Model Checking (A)
Algorithms (A)
Machine Learning (B+)
Computer Vision (B)
Artificial Intelligence (B+)
Database Management (O)

Introduction to C Programming (A)
Data Structures & Algorithms (B+)
Group Theory^{*}
Modern Cryptography^{*}
Information Theory & Coding^{*}
Computer Organization^{*}

RESEARCH PROJECTS



On Robust Coloring of Graphs and ETH

July 2025 – Present

Mentor: [Dr. Prafullkumar Tale](#)

- Studing a relaxation of proper coloring in which edges with same color are associated with some cost.
- Studied the EXPONENTIAL TIME HYPOTHESIS, and ETH-Preserving reductions.
- Investigated treewidth; applied tree decompositions to dynamic programming on hard problems.



Scientific Tool for Bridging Model Checking Ecosystems

Dec 2024 – Present

Mentor: [Dr. Arpit Sharma](#), PhD. [Shonak Shaha](#)

- The Model Checking ecosystem remains fragmented, with various tools and frameworks lacking seamless interoperability specifically between action based and state based model checking.
- Developing high-performance converters between action-labeled models ([CADP](#)) and ([mCRL2](#)) and state-labeled models ([PRISM](#)) / ([Storm](#)) to enable cross-ecosystem model checking.
- Emphasis on parser design, memory-efficient graph transformations, semantic preservation, and interoperability testing.



On the feasibility of parameterized algorithms for VC Dimensions

May 2025 – July 2025

Mentor: [Dr. Prafullkumar Tale](#)

- VC DIMENSION of a classification model is related to how complicated it can be, specifically in terms of the model's capacity to fit various datasets. Precisely, model's ability to generalize to unseen data.
- Tested efficient algorithms for computing the GRAPH VC DIMENSION.
- Assessed and compared the parameterized algorithm with current state-of-the-art for GRAPH VC DIMENSION for real world networks.



Reading Project on Graph Theory

Jan 2024 – Apr 2024

Mentor: [Dr. Prafullkumar Tale](#)

- Studied *A First Look at Graph Theory* (Clark & Holton) with problem solving.
- Reinforced concepts: Graphs & their types, connectivity, traversals, matching problem, planarity.

SKILLS

Python (NumPy, Pandas, Matplotlib, NetworkX) • C++ (CMake, Large Scale Projects) • Git & GitHub • LaTeX • Linux • Development automation (CI/CD with GitHub Actions) • Basic React

^{*}Ongoing

TECHNICAL PROJECTS

-  **Nifty Network Analysis & Visualization Dashboard** Oct 2025 – Nov 2025

 - Analyzed relationships among major Nifty companies and summarized insights in an interactive dashboard and formal report.
 - Delivered clear visual findings on market structure, influential firms, and cluster patterns for course project.
-  **Extensively Customizable Exam Scheduler using Graph Coloring** Oct 2024 – Jan 2025
Vivek Kumar, Rahul Jana, Ayushman Shaha, [Dr. Prafullkumar Tale](#)
 - Discovered that exam schedules for 2,000+ students in my college were being created manually, requiring significant effort.
 - Developed an engine that assigns exams to time slots & halls minimizing same-day conflicts ($\leq 24h$) using graph coloring + randomized optimization heuristics.
 - Supports multi-hall allocation when enrollment exceeds single capacity; leverages NetworkX for constraint modeling.
-  **Texture Classification & Face Clustering for Image Search** Sep 2024 – Nov 2024
 - The project aims to address the common challenge of navigating through directories containing a large collection of images, enabling users to efficiently filter and search for images.
 - Pipeline: MTCNN detection → InceptionResNetV1 embeddings → cosine similarity for face grouping.
 - Implemented batching to control GPU memory and multiple texture descriptors for retrieval precision.
-  **GUI Developer Intern** Feb 2024 – Aug 2024
Mentor: [Prof. Vardharajan Srinivasan](#) AITG Labs

 - Led integration of scientific engines ([Octopus](#), [GPAW](#), [NWChem](#)) within a Python toolkit for photo-induced phenomena simulations.
 - Automated submission workflows and job orchestration on national HPC clusters (PARAM-Ganga / PARAM-Kamrupa).
-  **Context aware Bag-of-Words Chatbot** Oct 2021 – Dec 2021
 - Built a context-aware chatbot supporting tasks like to-do lists, word definitions, note taking, and time queries across timezones.
 - Trained a sequential neural network in TensorFlow on a custom dataset tailored to project requirements.
-  **General Purpose Discord Bot** Mar 2021
 - Python project that utilized PostgreSQL to implement feature reach Discord bot.

TALKS & TEACHING

- Teaching Assistant - Theory of Computation** Aug 2025 – Dec 2025
Instructed by [Dr. Arpit Sharma](#)
-  **Fast Matrix Multiplication Algorithms** 1 Sept 2023
Math Club @ IISER Bhopal
 - Delivered a talk on Strassen's algorithm and its implications on computational complexity.

LEADERSHIP & ENGAGEMENT

- Club Coordinator** May 2024 – May 2025
Coding club @ IISER Bhopal
- Hackathon Organizer – Armacode 0** Jan 2024 – Apr 2024
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ADDITIONAL INTERESTS

Music (Guitar) • Competitive Programming • Speed Typing • Mathematical Visualization (Desmos, Manim)